



Tried ZFS on Linux?

ZFS is a revolutionary filesystem from Solaris with features miles ahead of ext3, the most-used filesystem under Linux. However, licence incompatibility makes it impossible to merge this open source filesystem to Linux. This article tells us how to use ZFS on FUSE, a tool that enables you to run ZFS on Linux—legally.

*J*eff Bonwick, the leader of the team at Sun Microsystems that developed ZFS, called it "...the last word in filesystems." It is indeed worthy of the praise considering its advanced yet easily maintainable features. ZFS, a pseudo-acronym for what was earlier called Zettabyte Filesystem, is a 128-bit filesystem, as opposed to the presently available 64-bits filesystems like ext4 and others.

Some of its excellent features include:

- **Simplified administration:** ZFS has a well-planned hierarchical structure with the uberblock (parent of all blocks) and disk label at the top, followed by pool-wide metadata, the filesystem's metadata, directories and files. The uberblock checksum is used as the digital signature for the entire filesystem. Besides property inheritance (utilising the hierarchical structure), ZFS provides auto management of mounting, sharing, compressions, ACLs, quotas and reservations, etc, making administration easier and more effective.

The filesystems in ZFS can be compared to directories in ordinary filesystems like ext3, and most administration tasks are done using just two commands—*zfs* and *zpool*.

- **Pooled storage:** ZFS has revolutionised the filesystem implementation and its management with the introduction of storage pools. Concepts like datasets (a generic term for volumes, filesystems, snapshots and clones) and pools (a large storage area available for the datasets) make filesystem handling easier for the administrator. Like the virtual memory model for a process, the filesystem can grow its usage space as required without any pre-determined space limits unless provided as 'quotas' within the pool model. 'Quotas' can be set, changed or removed at will. Also, a minimum 'reservation' space for each filesystem can be specified. One important aspect of the storage pool is the removal of volume management architecture, thus reducing a lot of complexity for the administrator.